CHAPTER C22
SEISMIC GROUND MOTION, LONG-PERIOD TRANSITION, AND RISK COEFFICIENT MAPS

Like the NEHRP Provisions (2009) and ASCE/SEI 7-10, this standard continues to use risk-targeted maximum considered earthquake (MCE_R) ground motion contour maps of 0.2-s and 1.0-s spectral response accelerations (Figs. 22-1 through 22-8), maximum considered earthquake geometric mean (MCE_G) peak ground acceleration maps (Figs. 22-9 through 22-13), and mapped risk coefficients at 0.2 s and 1 s (Figs. 22-18 and 22-19). However, the basis for these mapped values for the conterminous United States has been updated by the U.S. Geological Survey (USGS), as described below.

Furthermore, consistent with the site-specific procedures of Section 21.2.1.2 of ASCE/SEI 7-10 and this standard, but unlike the MCE_R ground motion maps of ASCE/SEI 7-10 (and unlike the site-specific procedures of the 2009 *NEHRP Recommended Provisions for Seismic Regulations for New Buildings and Other Structures*), the logarithmic standard deviation of the collapse fragility used in determining the mapped MCE_R values for the conterminous United States has also been updated (from 0.8 to 0.6). Although this standard also continues to use mapped long-period transition periods (Figs. 22-14 through 22-17), these mapped values have not been updated. Even for the conterminous United States, no significant changes are expected to the deaggregation computations that underlie the mapped long-period transition periods in ASCE/SEI 7-10.

The MCE_R ground motion, MCE_G peak ground acceleration, and risk coefficient maps incorporate the latest seismic hazard models developed by the USGS for the U.S. National Seismic Hazard Maps, including the latest seismic, geologic, and geodetic information on earthquake rates and associated ground shaking.

For the conterminous United States, the latest USGS model is documented in Petersen et al. (2013, 2014). This 2014 model supersedes versions released in 1996, 2002, and 2008. The most significant changes for the 2014 model fall into four categories, as follows:

1. For Central and Eastern United States (CEUS) sources:
   - Developed a moment magnitude-based earthquake catalog through 2012, replacing the 2008 m_b-based catalog;
   - Updated earthquake catalog completeness estimates, catalog of statistical parameters, treatment of nontectonic seismicity, and treatment of magnitude uncertainty;
   - Updated the distribution for maximum magnitude ($M_{\max}$) for background earthquakes based on a new analysis of global earthquakes in stable continental regions;
   - Updated the zonation for maximum magnitude, keeping the two-zone model that distinguishes craton and margin zones used in previous maps, and added a new four-zone model based on the Central and Eastern U.S. Seismic Source Characterization for Nuclear Facilities Project (CEUS-SSCn 2012) delineating the craton, Paleozoic margin, Mesozoic margin, and Gulf Coast;
   - Updated the smoothing algorithms for background seismicity, keeping the previous fixed-length Gaussian smoothing model, and adding a nearest-neighbor-type adaptive smoothing model;
   - Updated the New Madrid source model, including fault geometry, recurrence rates of large earthquakes, and alternative magnitudes from M6.6–M8.0 (keeping the highest weight about M7.5);
   - Adapted seismic sources such as Charleston, Wabash, Charlevoix, Commerce lineament, East Rift Margin, Mariana based on the CEUS-SSCn (2012) model; and
   - Updated the treatment of earthquakes that are potentially induced by underground fluid injection.
2. For Intermountain West and Pacific Northwest crustal sources:
   - Considered recommendations from the Basin and Range Province Earthquake Working Group on magnitude-frequency distributions for fault sources, smoothing parameters, comparison of historical and modeled seismicity rates, treatment of magnitude uncertainty, assessment of maximum magnitude, modeling of antithetic fault pairs, slip rate uncertainties, and dip uncertainty for normal faults (Lund 2012);
   - Updated the earthquake catalog and treatment of magnitude uncertainty in rate calculations;
   - Incorporated dips for normal faults of 35°, 50°, and 65° but applied the fault earthquake rate using only the 50° dip to the three alternatives;
   - Updated fault parameters for faults in Utah based on new data sets and models supplied by the Utah Geological Survey and the Working Group on Utah Earthquake Probabilities;
   - Introduced new combined geologic and geodetic inversion models for assessing fault slip rates on fault sources;
   - Implemented new models for Cascadia earthquake-rupture geometries and rates based on onshore (paleotsunami) and offshore (turbidite) studies;
   - Updated the model for deep (intraslab) earthquakes along the coasts of Oregon and Washington, including a new depth distribution for intraslab earthquakes;
   - Allowed for an M_{max} up to M8.0 for crustal and intraslab earthquakes; and
   - Added the Tacoma fault source and updated the South Whidbey Island fault source in Washington.
3. For California sources:
   - The USGS worked in cooperation with the Southern California Earthquake Center and the state of California to develop a new seismic source model based on the Uniform California Earthquake Rupture Forecast, Version 3 (WGCEP 2013) and new earthquake forecasts for

California, which include many more multisegment ruptures than in previous editions of the maps. These models were developed over the past several years and involved a major update of the methodology for calculating earthquake recurrence.

4. For ground motion models (or "attenuation relations"):
   - Included new earthquake ground motion models for active shallow crustal earthquakes (NGA-West 2) and subduction zone-related interface and intraslab earthquakes;
   - Adjusted the additional epistemic uncertainty model to account for regional variability and data availability;
   - Updated ground motion prediction equation weights using a new residual analysis based on the Next Generation Attenuation (NGA)-East ground motion database, reevaluated model weights in light of a preliminary Electric Power Research Institute (EPRI 2013) ground motion study, and included newly published ground motion prediction equations for stable continental regions;
   - Incorporated new and evaluated older ground motion models: five equations for the Western United States (WUS), nine for the CEUS, and four for the subduction interface and intraslab earthquakes; and
   - Increased the maximum distance from 200 km (124.3 mi) to 300 km (186.4 mi) when calculating ground motion from WUS crustal sources.

The 2014 updated National Seismic Hazard Maps differ from the 2008 maps in a variety of ways. The new ground motions vary locally depending on complicated changes in the underlying models. In the CEUS, the new earthquake catalog, completeness models, smoothing algorithms, magnitude uncertainty adjustments, and fault models increase the hazard in some places, and the new ground motion model-weighting scheme generally lowers the ground motions. The resulting maps for the CEUS can differ by $\pm 20\%$ compared to the 2008 maps because of interactions between the various parts of models summarized in the bullets above. In the Intermountain West region, the combined geologic and geodetic inversion models increase the hazard along the Wasatch fault and central Nevada region, but the new NGA-West2 ground motions tend to lower the hazard on the hanging walls of normal faults with respect to the 2008 maps. These counteracting effects can result in complicated patterns of changes. In the Pacific Northwest, the new Cascadia source model causes the hazard to increase by up to 40% in the southern Cascadia subduction zone because of the addition of possible M8 and greater earthquakes, but the model causes the hazard to decrease slightly along the northern Cascadia subduction zone because of reduced earthquake rates relative to the 2008 USGS hazard model. Subduction ground motions from the new models fall off faster with distance than motions in previous models, but they also tend to be higher near fault ruptures. In California, the new UCERF3 model (WGCEP 2013) accounts for earthquakes that rupture multiple faults, yielding larger magnitudes than applied in the previous model, but with smaller recurrence rates. However, they also include new ground motion models for strike-slip earthquakes, new slip rates from combined geodetic-geologic inversions, new faults, and an adaptive smoothing seismicity model that can locally increase the hazard compared to the previous model. At a specific site, it is important to examine all model changes, documented in Petersen et al. (2013) and Petersen (2014), to determine why the ground motions may have increased or decreased.

The combined effects of the updates to the USGS hazard model and to the collapse-fragility logarithmic standard deviation (or "beta value") on the $\mathrm{MCE_R}$ and $\mathrm{MCE_G}$ ground motion maps are demonstrated in Tables C22-2 through C22-4 for the same 34 locations considered in the 2009 and 2015 *NEHRP Provisions* listed in Table C22-1. In the tables, the $\mathrm{MCE_R}$ (S_S and S_1) and $\mathrm{MCE_G}$ (PGA) ground motions from the proposed maps are compared with those from ASCE/SEI 7-10 (and their equivalents from the 1997 Uniform Building Code). Furthermore, the updated site coefficients of this standard and those of ASCE/SEI 7-10 are applied to the corresponding mapped values to provide examples of the *design* spectral response accelerations (S_{DS} and S_{D1}) and *site-adjusted* peak ground accelerations (PGA_M) for an undetermined site class (Site Class D in ASCE/SEI 7-10, the worst case of Site Classes C and D in this standard). Lastly, the seismic design categories (SDCs) corresponding to the design spectral response accelerations are also compared in the tables.

It is important to bear in mind that the updated S_{DS}, S_{D1}, and PGA_M values in the ASCE/SEI 7-16 columns of the tables and the updated SDCs include the approved changes to the site coefficients, an up to 20% increase for an undetermined site class. Nevertheless, from Tables C22-2 and C22-3, it is apparent that the more severe SDC from S_{DS} and S_{D1} does not change for all but two of the 34 locations. The exceptions are the locations in Los Angeles and San Mateo, where the SDC based on S_{D1} (and S_1) alone decreases from E to D. For the Las Vegas location, the SDC based on S_{DS} alone increases from C to D, but the SDC based on S_{D1} was already D in ASCE/SEI 7-10. Close examination of the S_S map reveals a few additional areas where the SDC based on S_{DS} alone increases relative to ASCE/SEI 7-10, most notably:

- In southeastern New Hampshire, central Virginia, and at the border between Tennessee and North Carolina, S_S increases because of (i) inclusion of a widely used adaptive (as opposed to fixed-length) algorithm for smoothing historical seismicity rates, which increases hazard in areas of clustered historical seismicity, and (ii) changes to historical earthquake magnitudes and their rates based on the Central and Eastern U.S. Seismic Source Characterization for Nuclear Facilities project (CEUS-SSCn 2012), funded by the U.S. Department of Energy, the Electric Power Research Institute, and the U.S. Nuclear Regulatory Commission.
- In southwestern Oklahoma, S_S increases because of inclusion of a much broader range of potential earthquake magnitudes and rates for the Meers fault, based on CEUS-SSCn (2012).

At the aforementioned 34 locations, the ASCE/SEI 7-16 PGA values of Table C22-4, which have only been affected by the updates to the USGS National Seismic Hazard Maps (not the updated beta value), are within $\pm 20\%$ of the respective ASCE/SEI 7-10 values, with the exceptions below. Recall that a 20% decrease is the most allowed when a site-specific hazard analysis is performed in accordance with Chapter 21.

- For the San Diego location, the increase in the PGA value (and S_S value discussed below) is due to a combination of the addition of more offshore faults, the consideration of geodetic (GPS) data, the inclusion of lower magnitude earthquakes that can contribute significantly to PGA values (and 0.2-s spectral response accelerations), and the increases in the ground motions for large-magnitude earthquakes near strike-slip faults from the updated NGA-West 2 attenuation relations.
- For Vallejo, the increase in the PGA value is primarily caused by lengthening of the West Napa fault based on

Table C22-1 Latitudes and Longitudes for which MCE_R and MCE_G Ground Motions from ASCE/SEI 7-16 and ASCE/SEI 7-10 are Compared in Tables C22-2 through C22-4

	City and Location of Site			County or Metropolitan Statistical Area	
Region	**Name**	**Latitude**	**Longitude**	**Name**	**Population**
Southern California	Los Angeles	34.05	−118.25	Los Angeles	9,948,081
	Century City	34.05	−118.40		
	Northridge	34.20	−118.55		
	Long Beach	33.80	−118.20		
	Irvine	33.65	−117.80	Orange	3,002,048
	Riverside	33.95	−117.40	Riverside	2,026,603
	San Bernardino	34.10	−117.30	San Bernardino	1,999,332
	San Luis Obispo	35.30	−120.65	San Luis Obispo	257,005
	San Diego	32.70	−117.15	San Diego	2,941,454
	Santa Barbara	34.45	−119.70	Santa Barbara	400,335
	Ventura	34.30	−119.30	Ventura	799,720
	Total Population—S. California		22,349,098	Population—8 Counties	21,374,788
Northern California	Oakland	37.80	−122.25	Alameda	1,502,759
	Concord	37.95	−122.00	Contra Costa	955,810
	Monterey	36.60	−121.90	Monterey	421,333
	Sacramento	38.60	−121.50	Sacramento	1,233,449
	San Francisco	37.75	−122.40	San Francisco	776,733
	San Mateo	37.55	−122.30	San Mateo	741,444
	San Jose	37.35	−121.90	Santa Clara	1,802,328
	Santa Cruz	36.95	−122 05	Santa Cruz	275,359
	Vallejo	38.10	−122.25	Solano	423,473
	Santa Rosa	38.45	−122.70	Sonoma	489,290
	Total Population—N. California		14,108,451	Population—10 Counties	8,621,978
Pacific Northwest	Seattle	47.60	−122.30	King, WA	1,826,732
	Tacoma	47.25	−122.45	Pierce, WA	766,878
	Everett	48.00	−122.20	Snohomish, WA	669.887
	Portland	45.50	−122.65	Portland Metro, OR (3)	1,523,690
	Total Population—OR and WA		10,096,556	Population—6 Counties	4,787,187
Other WUS	Salt Lake City	40.75	−111.90	Salt Lake, UT	978,701
	Boise	43.60	−116.20	Ada/Canyon, ID (2)	532,337
	Reno	39.55	−119.80	Washoe, NV	396,428
	Las Vegas	36.20	−115.15	Clarke, NV	1,777,539
	Total Population—ID/UT/NV		6,512,057	Population—5 Counties	3,685,005
CEUS	St. Louis	38.60	−90.20	St. Louis MSA (16)	2,786,728
	Memphis	35.15	−90.05	Memphis MSA (8)	1,269,108
	Charleston	32.80	−79.95	Charleston MSA (3)	603,178
	Chicago	41.85	−87.65	Chicago MSA (7)	9,505,748
	New York	40.75	−74.00	New York MSA (23)	18,747,320
	Total Population—MO/TN/SC/IL/NY		48,340,918	Population—57 Counties	32,912,082

Note: The 34 locations come from the 2009 and 2015 *NEHRP Provisions*. It is important to note that these locations are each just one of many in the named cities, and their ground motions may be significantly different than those at other locations in the cities.

the Statewide Community Fault Model (see WGCEP 2013).

- For Reno, the increase in the PGA value is primarily caused by the NGA-West 2 attenuation relations. Note that the NGA-West 2 attenuation relations are based on double the strong-motion data used for the NGA-West 1 relations.
- For Las Vegas, the increase in the PGA value (and S_S and S_1 values discussed below) is primarily caused by an increase in the estimated rate of earthquakes on the Eglington fault, based on recent studies and a recommendation from the state geologist of Nevada.
- For Memphis, the increase in the PGA value is caused by consideration of an alternative model for the New Madrid Seismic Zone based on the aforementioned CEUS-SSCn reference (2012).
- For Charleston, the increase in the PGA value (and S_S value discussed below) is caused by reevaluation of the data from the Charleston earthquakes and consequent revision of the Charleston seismic source model (CEUS-SSCn 2012). The USGS adopted this revised model based on its own analysis, as well as the recommendations of its steering committee and participants of a regional workshop.

The ASCE/SEI 7-16 S_S values of Table C22-2 have been affected by the update to the ASCE/SEI 7-10 beta value (ground motion changes of up to approximately 10%), in addition to the updated USGS National Seismic Hazard Maps. The only locations where the S_S values have changed by more than 20% with respect to ASCE/SEI 7-10 are San Diego and Santa Barbara, California; Las Vegas, Nevada; and Charleston, South Carolina. Please see the explanation below of the changes in the USGS

Table C22-2 A Comparison of the Short-Period Design Spectral Response Accelerations (S_{DS} Values) from this Standard and ASCE/SEI 7-10 and Their Equivalents from the 1997 Uniform Building Code for the 34 Locations Considered in the 2009 and 2015 *Provisions*

		1997 UBC		ASCE/SEI 7-10			ASCE/SEI 7-16		
Region	Location Name	Zone	2.5*C_a	S_S (g)	S_{DS} (g)*	SDC_S**	S_S (g)	S_{DS} (g)***	SDC_S**
Southern California	Los Angeles	4	1.10	2.40	1.60	D	1.97	1.58	D
	Century City	4 (NF)	1.32	2.16	1.44	D	2.11	1.69	D
	Northridge	4	1.10	1.69	1.13	D	1.74	1.39	D
	Long Beach	4 (NF)	1.43	1.64	1.10	D	1.68	1.35	D
	Irvine	4	1.10	1.55	1.03	D	1.25	1.00	D
	Riverside	4	1.10	1.50	1.00	D	1.50	1.20	D
	San Bernardino	4 (NF)	1.32	2.37	1.58	D	2.33	1.86	D
	San Luis Obispo	4	1.10	1.12	0.78	D	1.09	0.87	D
	San Diego	4 (NF)	1.43	1.25	0.84	D	1.58	1.26	D
	Santa Barbara	4 (NF)	1.43	2.83	1.89	D	2.12	1.70	D
	Ventura	4 (NF)	1.43	2.38	1.59	D	2.02	1.62	D
	Weighted Mean		**1.25**	**1.83**	**1.22**		**1.75**	**1.40**	
Northern California	Oakland	4 (NF)	1.43	1.86	1.24	D	1.88	1.51	D
	Concord	4	1.10	2.08	1.38	D	2.22	1.78	D
	Monterey	4	1.10	1.53	1.02	D	1.33	1.06	D
	Sacramento	3	0.90	0.67	0.57	D	0.57	0.51	D
	San Francisco	4	1.10	1.50	1.00	D	1.50	1.20	D
	San Mateo	4 (NF)	1.28	1.85	1.23	D	1.80	1.44	D
	San Jose	4	1.10	1.50	1.00	D	1.50	1.20	D
	Santa Cruz	4	1.10	1.52	1.01	D	1.59	1.27	D
	Vallejo	4 (NF)	1.19	1.50	1.00	D	1.50	1.20	D
	Santa Rosa	4 (NF)	1.65	2.51	1.67	D	2.41	1.93	D
	Weighted Mean		**1.18**	**1.60**	**1.08**		**1.59**	**1.28**	
Pacific Northwest	Seattle			1.36	0.91	D	1.40	1.12	D
	Tacoma			1.30	0.86	D	1.36	1.08	D
	Everett			1.27	0.85	D	1.20	0.96	D
	Portland			0.98	0.72	D	0.89	0.71	D
	Weighted Mean			**1.22**	**0.83**		**1.20**	**0.96**	
Other WUS	Salt Lake City			1.54	1.03	D	1.54	1.24	D
	Boise			0.31	0.32	B	0.31	0.32	B
	Reno			1.50	1.00	D	1.47	1.17	D
	Las Vegas			0.49	0.46	C	0.65	0.55	D
	Weighted Mean			**0.85**	**0.65**		**0.92**	**0.77**	
CEUS	St. Louis			0.44	0.42	C	0.46	0.44	C
	Memphis			1.01	0.74	D	1.02	0.82	D
	Charleston			1.15	0.80	D	1.42	1.13	D
	Chicago			0.13	0.14	A	0.12	0.13	A
	New York			0.28	0.29	B	0.29	0.30	B
	Weighted Mean			**0.30**	**0.29**		**0.30**	**0.30**	

Note: It is important to bear in mind that the design spectral response accelerations (S_{DS} values) and the seismic design categories in the table include the effects of the updated site coefficients of this standard.
*The ASCE/SEI 7-10 S_{DS} values are calculated using the ASCE/SEI 7-10 F_a site coefficients, for an undetermined site class (assigned Site Class D in ASCE/SEI 7-10).
**The ASCE/SEI 7-16 S_{DS} values are calculated using the updated ASCE/SEI 7-16 F_a site coefficients, also for an undetermined site class (assigned the worst case of Site Classes C and D in ASCE/SEI 7-16).
***The SDC_S categories corresponding to the S_{DS} values (and Risk Category I/II/III) are assigned using Table 11.6-1 and the SDC E definition (of ASCE/SEI 7-10 and 7-16) alone.

National Seismic Hazard Maps at the Santa Barbara location and the explanations stated previously for the San Diego, Las Vegas, and Charleston locations. For the other locations explained previously—Vallejo, California; Reno, Nevada; and Memphis, Tennessee—the S_S values have changed by at most 2%.

- For the Santa Barbara location, the decrease in the S_1 value is a combination of the decrease in ground motions over reverse faults from the NGA-West 2 attenuation relations and the fact that more multifault earthquakes have been allowed in UCERF3 (WGCEP 2013), relative to the hazard model underlying the ASCE/SEI 7-10 ground motion maps. This, in effect, lowers the rate of earthquakes and, hence, lowers the probabilistic ground motions.

The ASCE/SEI 7-16 S_1 values of Table C22-3 have also been affected by the update to the ASCE/SEI 7-10 beta value (changes of up to approximately 10%) and the updated USGS National Seismic Hazard Maps. The only locations where the S_1 values have changed by more than 20% with respect to ASCE/SEI 7-10 are Irvine, Santa Barbara, and Las Vegas. Please see the explanation below of the changes in the USGS National Seismic Hazard Maps at the Irvine, California, location and the previous explanations for the Santa Barbara and Las Vegas

Table C22-3 A Comparison of the 1.0-s MCE_R Design Spectral Response Accelerations (S_{D1} values) from this Standard and ASCE/SEI 7-10 and Their Equivalents from the 1997 Uniform Building Code for the 34 Locations Considered in the 2009 and 2015 *Provisions*

		1997 UBC		ASCE/SEI 7-10			ASCE/SEI 7-16		
Region	Location Name	Zone	2.5^*C_a	S_1 (g)	S_{D1} (g)*	SDC_1***	S_1 (g)	S_{D1} (g)**	SDC_1***
Southern California	Los Angeles	4 (NF)	0.72	0.84	0.84	E	0.70	0.79	D
	Century City	4 (NF)	0.93	0.80	0.80	E	0.75	0.85	E
	Northridge	4	0.64	0.60	0.60	D	0.60	0.68	D
	Long Beach	4 (NF)	1.02	0.62	0.62	D	0.61	0.69	D
	Irvine	4	0.64	0.57	0.57	D	0.45	0.55	D
	Riverside	4	0.64	0.60	0.60	D	0.58	0.67	D
	San Bernardino	4 (NF)	0.93	1.08	1.08	E	0.93	1.06	E
	San Luis Obispo	4 (NF)	0.77	0.43	0.45	D	0.40	0.51	D
	San Diego	4 (NF)	1.02	0.48	0.49	D	0.53	0.62	D
	Santa Barbara	4 (NF)	1.02	0.99	0.99	E	0.77	0.88	E
	Ventura	4 (NF)	1.02	0.90	0.90	E	0.76	0.86	E
	Weighted Mean		**0.83**	**0.70**	**0.70**		**0.63**	**0.73**	
Northern California	Oakland	4 (NF)	1.04	0.75	0.75	D	0.72	0.81	D
	Concord	4 (NF)	0.77	0.73	0.73	D	0.67	0.76	D
	Monterey	4 (NF)	0.77	0.56	0.56	D	0.50	0.60	D
	Sacramento	3	0.54	0.29	0.35	D	0.25	0.35	D
	San Francisco	4 (NF)	0.74	0.64	0.64	D	0.60	0.68	D
	San Mateo	4 (NF)	0.95	0.86	0.86	E	0.74	0.83	D
	San Jose	4 (NF)	0.69	0.60	0.60	D	0.60	0.68	D
	Santa Cruz	4 (NF)	0.72	0.60	0.60	D	0.60	0.68	D
	Vallejo	4 (NF)	0.87	0.60	0.60	D	0.60	0.68	D
	Santa Rosa	4 (NF)	1.28	1.04	1.04	E	0.94	1.06	E
	Weighted Mean		**0.81**	**0.65**	**0.65**		**0.61**	**0.70**	
Pacific Northwest	Seattle			0.53	0.53	D	0.49	0.59	D
	Tacoma			0.51	0.51	D	0.47	0.57	D
	Everett			0.48	0.49	D	0.43	0.53	D
	Portland			0.42	0.44	D	0.39	0.50	D
	Weighted Mean			**0.48**	**0.49**		**0.45**	**0.55**	
Other WUS	Salt Lake City			0.56	0.56	D	0.55	0.65	D
	Boise			0.11	0.17	C	0.11	0.17	C
	Reno			0.52	0.52	D	0.52	0.61	D
	Las Vegas			0.17	0.24	D	0.21	0.30	D
	Weighted Mean			**0.30**	**0.34**		**0.32**	**0.41**	
CEUS	St. Louis			0.17	0.24	D	0.16	0.25	D
	Memphis			0.35	0.40	D	0.35	0.45	D
	Charleston			0.37	0.41	D	0.41	0.52	D
	Chicago			0.06	0.10	B	0.06	0.10	B
	New York			0.07	0.11	B	0.06	0.10	B
	Weighted Mean			**0.09**	**0.14**		**0.09**	**0.13**	

Note: It is important to bear in mind that the design spectral response accelerations (S_{D1} values) and the seismic design categories in the table include the effects of the updated site coefficients of this standard.
*The ASCE/SEI 7-10 S_{D1} values are calculated using the ASCE/SEI 7-10 F_v site coefficients, for an undetermined site class (assigned Site Class D in ASCE/SEI 7-10).
**The ASCE/SEI 7-16 S_{D1} values are calculated using the updated ASCE/SEI 7-16 F_v site coefficients, also for an undetermined site class (assigned the worst case of Site Classes C and D in ASCE/SEI 7-16).
***The SDC_1 categories corresponding to the S_{D1} values (and Risk Category I/II/III) are assigned using Table 11.6-2 and the SDC E definition (of ASCE/SEI 7-10 and 7-16) alone.

locations. For the other locations explained previously—San Diego, Vallejo, Reno, Memphis, and Charleston—the S_1 values have changed by at most 13%.

- For the Irvine location, the decrease in the S_1 values is caused primarily by a decrease in ground motions over reverse faults from the NGA-West 2 attenuation relations, and secondarily to the allowance for more multifault earthquakes in UCERF 3 (WGCEP 2013) that is described in the preceding bullet.

In summary, with the updates to the USGS National Seismic Hazard maps for the conterminous United States and the updated logarithmic standard deviation of the collapse fragilities, (i) the Seismic Design Categories for 32 of the 34 Table C22-1 locations do not change with respect to ASCE/SEI 7-10; (ii) the PGA values change by −15% to +17% for 28 of the 34 locations; (iii) the S_S values change by −19% to +7 for 30 of the locations; and (iv) the S_1 values change by −17% to +13% for 31 of the locations.

Like previous versions of the USGS national seismic hazard model, the 2014 model purposefully excludes swarms of earthquakes that may be causally related to industrial fluid processes, such as hydrocarbon production or wastewater disposal. The excluded swarms are identified in Figure 15 of Petersen (2014). Whereas an average of 21 earthquakes per year of magnitude greater than 3 occurred from 1967 to 2000 in the CEUS, more

Table C22-4 A Comparison of the MCE_G Peak Ground Accelerations (PGA Values) from this Standard and ASCE/SEI 7-10 for the 34 Locations Considered in the 2009 and 2015 *Provisions*

		ASCE/SEI 7-10		ASCE/SEI 7-16	
Region	Location Name	PGA (g)	PGA_M (g)*	PGA (g)	PGA_M (g)**
Southern California	Los Angeles	0.91	0.91	0.84	1.01
	Century City	0.81	0.81	0.91	1.09
	Northridge	0.62	0.62	0.71	0.86
	Long Beach	0.64	0.64	0.74	0.89
	Irvine	0.60	0.60	0.53	0.63
	Riverside	0.50	0.50	0.50	0.60
	San Bernardino	0.91	0.91	0.98	1.18
	San Luis Obispo	0.44	0.47	0.48	0.58
	San Diego	0.57	0.57	0.72	0.86
	Santa Barbara	1.09	1.09	0.93	1.11
	Ventura	0.91	0.91	0.88	1.06
	Weighted Mean	**0.70**	**0.70**	**0.74**	**0.89**
Northern California	Oakland	0.72	0.72	0.79	0.95
	Concord	0.79	0.79	0.90	1.07
	Monterey	0.59	0.59	0.58	0.69
	Sacramento	0.23	0.31	0.24	0.32
	San Francisco	0.57	0.57	0.58	0.70
	San Mateo	0.73	0.73	0.78	0.93
	San Jose	0.50	0.50	0.57	0.69
	Santa Cruz	0.59	0.59	0.67	0.81
	Vallejo	0.51	0.51	0.62	0.74
	Santa Rosa	0.97	0.97	1.02	1.22
	Weighted Mean	**0.59**	**0.60**	**0.65**	**0.78**
Pacific Northwest	Seattle	0.56	0.56	0.60	0.72
	Tacoma	0.50	0.50	0.50	0.60
	Everett	0.52	0.52	0.51	0.62
	Portland	0.42	0.46	0.40	0.48
	Weighted Mean	**0.50**	**0.51**	**0.51**	**0.61**
Other WUS	Salt Lake City	0.67	0.67	0.70	0.84
	Boise	0.12	0.19	0.14	0.21
	Reno	0.50	0.50	0.62	0.74
	Las Vegas	0.20	0.28	0.28	0.37
	Weighted Mean	**0.34**	**0.39**	**0.41**	**0.51**
CEUS	St. Louis	0.23	0.31	0.27	0.36
	Memphis	0.50	0.50	0.61	0.73
	Charleston	0.75	0.75	0.93	1.12
	Chicago	0.07	0.11	0.06	0.09
	New York	0.17	0.25	0.18	0.26
	Weighted Mean	**0.17**	**0.23**	**0.18**	**0.25**

Note: It is important to bear in mind that the site-adjusted peak ground accelerations (PGA_M values) in the table include the effects of the updated site coefficients of this standard.
*The ASCE/SEI 7-10 PGA_M values are calculated using the ASCE/SEI 7-10 F_{PGA} site coefficients, for an undetermined site class (assigned Site Class D in ASCE/SEI 7-10).
**The ASCE/SEI 7-16 PGA_M values are calculated using the updated ASCE/SEI 7-16 F_{PGA} site coefficients, also for an undetermined site class (assigned the worst case of Site Classes C and D in ASCE/SEI 7-16).

than 300 such earthquakes have occurred from 2010 through 2012. Thus, in the areas of the excluded swarms, the seismic hazard might be higher than that estimated by the 2014 USGS model; on the other hand, it could decrease significantly in the coming years with changes in the fluid processes. Treatment of the potentially induced earthquake swarms in hazard modeling is a topic of active research.

In 2012, the USGS developed seismic hazard models for Guam and the Northern Mariana Islands (Guam/NMI) and for American Samoa using the same type of seismic hazard analysis that underlies the 2008 model for the conterminous United States. The hazard models for the islands are documented in Mueller et al. (2012) and Petersen et al. (2012), respectively. In comparing the MCE_R ground motion maps derived from these USGS hazard models to the geographically constant values stipulated for Guam and American Samoa (Tutuila) in the 2010 and previous editions of ASCE/SEI 7, it is important to bear in mind that the latter were not computed via seismic hazard modeling. According to the commentary of the NEHRP Provisions (1997), the geographically constant values were merely conversions, via rough approximations, from values on the NEHRP Provisions (1994) maps that had been in use for nearly 20 years. As such, they did not take into account the 1993 Guam earthquake that was the largest ever recorded in the region and caused considerable damage, the 2009 earthquake near American Samoa that caused a tsunami, nor the 2008 "Next Generation Attenuation (NGA)" and another 2006 empirical ground motion prediction equation that have now been used for both Guam/NMI and American Samoa. This and other such information is directly used in the seismic hazard modeling that is the basis for the MCE_R ground motion, MCE_G peak ground acceleration, and risk coefficient maps for Guam/NMI and American Samoa in this standard.

RISK-TARGETED MAXIMUM CONSIDERED EARTHQUAKE (MCE_R) GROUND MOTION MAPS

As introduced in the NEHRP Provisions (2009) and ASCE/SEI 7-10, the MCE_R ground motion maps are derived from underlying USGS seismic hazard models in a manner that is significantly different from that of the mapped values of MCE ground motions in previous editions of the NEHRP Provisions and ASCE/SEI 7. These differences include use of (1) probabilistic ground motions that are risk-targeted, rather than uniform hazard, (2) deterministic ground motions that are based on the 84th percentile (approximately 1.8 times median), rather than 1.5 times median response spectral acceleration for sites near active faults, and (3) ground motion intensity that is based on maximum, rather than the average (geometrical mean), response spectra acceleration in the horizontal plane.

The MCE_R ground motion maps have been prepared in accordance with the site-specific procedures of Section 21.2. More specifically, they represent the lesser of probabilistic ground motions defined in Section 21.2.1 and deterministic ground motions defined in Section 21.2.2, in accordance with Section 21.2.3. The preparation of the probabilistic and deterministic ground motions is described below.

The probabilistic ground motions have been calculated using Method 2 of Section 21.2.1 and the latest USGS hazard curves (of mean annual frequency of exceedance versus ground motion level) computed in accordance with Section 21.2 at gridded locations covering the United States and its territories. The USGS hazard curves are first converted from geometric-mean ground motions (output by the ground motion attenuation relations available to the USGS) to ground motions in the maximum direction of horizontal spectral response acceleration, with one exception. The USGS hazard curves for Hawaii, without conversion, are deemed to represent the maximum-response ground motions because of the attenuation relations applied there. For the other regions, the conversions were done by applying the factors specified in the site-specific procedures (Section 21.2) of ASCE/SEI 7-10 and this standard, namely 1.1 at 0.2 s and 1.3 at 1.0 s. The collapse fragilities used in calculating the probabilistic ground motions have a logarithmic standard deviation (or "beta value") of 0.6, as specified in ASCE/SEI 7-10 and this standard (Section 21.2.1), for the conterminous United States, Guam and the Northern Mariana Islands, and American Samoa. For the other regions (Hawaii, Puerto Rico and the U.S. Virgin Islands, and Alaska), where the latest USGS hazard curves predate the change of the logarithmic standard deviation from the NEHRP Provisions (2009) to ASCE/SEI 7-10, the beta value is 0.8. Please see Luco et al. (2007) for more information on the development of risk-targeted probabilistic ground motions.

The deterministic ground motions have been calculated using the "characteristic earthquakes on all known active faults" (quoted from Section 21.2.2) that the USGS uses in computing the probabilistic hazard curves. The largest characteristic magnitude considered by the USGS on each fault, excluding any lower weighted magnitudes from the USGS logic tree for epistemic uncertainty, is used for the deterministic ground motions. The active faults considered for the deterministic ground motions are those that have evidence of slip during Holocene time (the past 12,000 years, approximately), plus those with reported geologic rates of slip larger than 0.0004 in./year (0.1 mm/year). This slip rate can result in a magnitude 7 earthquake, which on average corresponds to 3.94 ft (1.2 m) of slip (Wells and Coppersmith 1994), over a 12,000-year time period; 0.0004 in./year (0.1 mm/year) also is the slip rate assigned by the Working Group on California Earthquake Probabilities (WGCEP 2013) to faults that, with the information available, could only be categorized as having a slip rate less than 0.0008 in./year (0.2 mm/year). At a user-input location, the fault (among hundreds) and corresponding magnitude that govern its deterministic ground motion is output by the USGS web tool briefly described in a section of this commentary below. For all the deterministic faults and magnitudes, the USGS has computed median (50th percentile), geometric-mean ground motions. To convert to maximum-response ground motions, the same scale factors described in the preceding paragraph for probabilistic ground motions are applied. To approximately convert to 84th percentile ground motions, the maximum-response ground motions are multiplied by 1.8.

MAXIMUM CONSIDERED EARTHQUAKE GEOMETRIC MEAN (MCE_G) PGA MAPS

Like the NEHRP Provisions (2009) and ASCE/SEI 7-10, but not previous editions, this standard includes contour maps of maximum considered earthquake geometric mean (MCE_G) peak ground acceleration, PGA, Figs. 22-9 through 22-13, for use in geotechnical investigations (Section 11.8.3). In contrast to the MCE_R ground motion maps, the maps of MCE_G PGA are defined in terms of geometric mean (rather than maximum direction) intensity and a 2% in 50-year hazard level (rather than a 1% in 50-year risk). Like the MCE_R ground motion maps, the maps of MCE_G PGA are governed near major active faults by deterministic values defined as 84th-percentile ground motions. The MCE_G PGA maps have been prepared in accordance with the site-specific procedures of Section 21.5 of ASCE/SEI 7-10 and this standard.

LONG-PERIOD TRANSITION MAPS

The maps of the long-period transition period, T_L (Figs. 22-14 through 22-17), were introduced in ASCE/SEI 7-05. They were prepared by the USGS in response to Building Seismic Safety Council recommendations and were subsequently included in the NEHRP Provisions (2003). See Section C11.4.6 for a discussion of the technical basis of these maps. The value of T_L obtained from these maps is used in Eq. (11.4-7) to determine values of S_a for periods greater than T_L.

The exception in Section 15.7.6.1, regarding the calculation of S_{ac}, the convective response spectral acceleration for tank response, is intended to provide the user the option of computing this acceleration with three different types of site-specific procedures: (a) the procedures in Chapter 21, provided that they cover the natural period band containing T_c, the fundamental convective period of the tank-fluid system; (b) ground motion simulation methods using seismological models; and (c) analysis of representative accelerogram data. Elaboration of these procedures is provided below.

With regard to the first procedure, attenuation equations have been developed for the western United States (Next Generation Attenuation, e.g., Power et al. 2008) and for the central and eastern United States (e.g., Somerville et al. 2001) that cover the period band, 0 to 10 s. Thus, for $T_c \leq 10$ s, the fundamental convective period range for nearly all storage tanks, these attenuation equations can be used in the same probabilistic seismic hazard analysis (PSHA) and deterministic seismic hazard analysis (DSHA) procedures described in Chapter 21 to compute S_a (T_c). The 1.5 factor in Eq. (15.7-11), which converts a 5% damped spectral acceleration to a 0.5% damped value, could then be applied to obtain S_{ac}. Alternatively, this factor could be established by statistical analysis of 0.5% damped and 5% damped response spectra of accelerograms representative of the ground motion expected at the site.

In some regions of the United States, such as the Pacific Northwest and southern Alaska, where subduction-zone earthquakes dominate the ground motion hazard, attenuation equations for these events only extend to periods between 3 and 5 s, depending on the equation. Thus, for tanks with T_c greater than these periods, other site-specific methods are required.

The second site-specific method to obtain S_a at long periods is simulation through the use of seismological models of fault rupture and wave propagation (e.g., Graves and Pitarka 2004, Hartzell and Heaton 1983, Hartzell et al. 1999, Liu et al. 2006, and Zeng et al. 1994). These models could range from simple seismic source-theory and wave-propagation models, which currently form the basis for many of the attenuation equations used in the central and eastern United States, for example, to more complex numerical models that incorporate finite fault rupture for scenario earthquakes and seismic wave propagation through 2D or 3D models of the regional geology, which may include basins. These models are particularly attractive for computing long-period ground motions from great earthquakes ($M_w \geq \sim 8$) because ground motion data are limited for these events. Furthermore, the models are more accurate for predicting longer period ground motions because (a) seismographic recordings may be used to calibrate these models and (b) the general nature of the 2D or 3D regional geology is typically fairly well resolved at these periods and can be much simpler than would be required for accurate prediction of shorter period motions.

A third site-specific method is the analysis of the response spectra of representative accelerograms that have accurately recorded long-period motions to periods greater than T_c. As T_c increases, the number of qualified records decreases. However, as digital accelerographs continue to replace analog accelerographs, more recordings with accurate long-period motions are becoming available. Nevertheless, a number of analog and digital recordings of large and great earthquakes are available that have accurate long-period motions to 8 s and beyond. Subsets of these records, representative of the earthquake(s) controlling the ground motion hazard at a site, can be selected. The 0.5% damped response spectra of the records can be scaled using seismic source theory to adjust them to the magnitude and distance of the controlling earthquake. The levels of the scaled response spectra at periods around T_c can be used to determine S_{ac}. If the subset of representative records is limited, then this method should be used in conjunction with the aforementioned simulation methods.

RISK COEFFICIENT MAPS

Like those in the NEHRP Provisions (2009) and ASCE/SEI 7-10 (where they were introduced), the risk coefficient maps in this standard (Figs. 22-18 and 22-19) provide factors, C_{RS} and C_{R1}, that are used in the site-specific procedures of Chapter 21 (Section 21.2.1.1, Method 1). These factors are implicit in the MCE_R ground motion maps.

The mapped risk coefficients are the ratios of (i) risk-targeted probabilistic ground motions (for 1% in 50 years collapse risk) derived from the USGS probabilistic seismic hazard curves, as described in the MCE_R ground motion maps section above, to (ii) corresponding uniform-hazard (2% in 50 years ground motion exceedance probability) ground motions that are simply interpolated from the USGS hazard curves. Note that these ratios (risk coefficients) are invariant to maximum-response scale factors that are applied to both the numerator and denominator.

GROUND MOTION WEB TOOL

The USGS has developed a companion web tool that calculates location-specific spectral values based on latitude and longitude. The calculated values are based on the gridded values used to prepare the maps. The spectral values can be adjusted for site class effects within the program using the site classification procedure in Section 20.1 and the site coefficients in Sections 11.4 and 11.8. The companion tool may be accessed at the USGS Earthquake Hazards Program website or through other hazards mapping tools. The tool should be used to establish spectral values for design because the maps found in this chapter are too small to provide accurate spectral values for many sites.

UNIFORM HAZARD AND DETERMINISTIC GROUND MOTION MAPS

Implicit in the MCE_R ground motion, MCE_G PGA, and risk coefficient maps provided are uniform hazard (2% in 50 years ground motion exceedance probability) and deterministic (84th percentile) ground motions. The NEHRP *Provisions* (2009) provided maps of such uniform hazard and deterministic ground motions, but *ASCE/SEI 7-10* and this standard do not. Instead, uniform hazard and deterministic ground motion maps consistent with this chapter are provided. Furthermore, values from these maps can be obtained via the ground motion software tool previously described.

It is important to note that the provided uniform hazard ground motion maps are for maximum direction of horizontal spectral response acceleration. As such, they are different than the maps of geometric mean spectral response acceleration provided elsewhere on the USGS Earthquake Hazards Program website. The provided deterministic ground motion maps are also for the maximum direction, but no geometric mean counterparts are provided. The USGS prepares the deterministic ground motion maps solely for the purposes of this standard, following the definition of deterministic ground motions in Section 21.2.2 (with the 84th percentile approximated as 1.8 times the median).

REFERENCES

CEUS-SSCn. (2012). Central and Eastern United States seismic source characterization for nuclear facilities: Electric Power Research Institute, U.S. Department of Energy and U.S. Nuclear Regulatory Commission. EPRI, Palo Alto, CA.

Electric Power Research Institute (EPRI). (2013). "EPRI (2004, 2006) Ground-Motion Model (GMM) Review Project," *EPRI Technical Report*, Product ID 3002000717, http://www.epri.com

Graves, R. W., and Pitarka, A. (2004). "Broadband time history simulation using a hybrid approach." Paper 1098, *Proc., 13th World Conference on Earthquake Engineering*, Vancouver.

Hartzell, S., and Heaton, T. (1983). "Inversion of strong ground motion and teleseismic waveform data for the fault rupture history of the 1979 Imperial Valley, California, earthquake." *Bull. Seismol. Soc. of Am.*, 73, 1553–1583.

Hartzell, S., Harmsen, S., Frankel, A., and Larsen, S. (1999). "Calculation of broadband time histories of ground motion: Comparison of methods and validation using strong ground motion from the 1994 Northridge earthquake." *Bull. Seismol. Soc. of Am.*, 89, 1484–1504.

Liu, P., Archuleta, R. J., and Hartzell, S. H. (2006). "Prediction of broadband ground-motion time histories: Hybrid low/high-frequency method with correlated random source parameters." *Bull. Seismol. Soc. of Am.*, 96, 2118–2130.

Luco, N., Ellingwood, B. R., Hamburger, R. O., Hooper, J. D., Kimball, J. K., and Kircher, C. A. (2007). "Risk-targeted vs. current seismic design maps for the conterminous United States." *Proc., Structural Engineers Association of California 76th Annual Convention*. SEAOC, Sacramento, CA.

Lund, W. R., ed. (2012). "Basin and range province earthquake working group II—Recommendations to the U.S. Geological Survey national seismic hazard mapping program for the 2014 update of the national seismic hazard maps: Utah Geological Survey." *USGS Open File Report 591*, Utah Geological Survey, Salt Lake City.

Mueller, C. S., Haller, K. M., Luco, N., Petersen, M. D., and Frankel, A. D. (2012). "Seismic hazard assessment for Guam and the Northern Marianna Islands," *USGS Open File Report 2012-1015*. USGS, Golden, CO.

National Earthquake Hazards Reduction Program (NEHRP). (1994). "Recommended provisions for seismic regulations for new buildings," *FEMA 222*. Building Seismic Safety Council, National Institute of Building Sciences, Washington, DC.

NEHRP. (1997). "Recommended provisions for seismic regulations for new buildings and other structures," *FEMA 302*. Building Seismic Safety Council, National Institute of Building Sciences, Washington, DC.

NEHRP. (2003). "Recommended provisions for seismic regulations for new buildings and other structures," *FEMA 450*. Building Seismic Safety Council, National Institute of Building Sciences, Washington, DC.

NEHRP. (2009). "Recommended provisions for seismic regulations for new buildings and other structures," *FEMA 750*. Building Seismic Safety Council, National Institute of Building Sciences, Washington, DC.

NEHRP. (2015). "Recommended provisions for seismic regulations for new buildings and other structures," *FEMA 1050*. Building Seismic Safety Council, National Institute of Building Sciences, Washington, DC.

Petersen, M.D. (2014). "Documentation for the 2014 update of the United States national seismic hazard maps," *USGS Open File Report 2014-1091*. USGS, Reston, VA.

Petersen, M. D., Moschetti, M. P., Powers, P. M., Mueller, C. S., Haller, K. M., Frankel, A. D., et al. (2013). "Documentation for the 2014 update of the United States National seismic hazard maps." USGS Administrative Report prepared for the Building Seismic Safety Council, Washington, DC.

Petersen, M. D., S. C., Harmsen, K. S., Rukstales, C. S., Mueller, D. E., McNamara, N., Luco, and M., Walling (2012). "Seismic Hazard of American Samoa and Neighboring South Pacific Islands: Data, Methods, Parameters, and Results," *USGS Open File Report* 2008–1087. USGS, Golden, CO.

Power, M., Chiou, B., Abrahamson, N., Bozorgnia, Y., Shantz, T., and Roblee, C. (2008). "An overview of the NGA project." *Earthquake Spectra Special Issue on the Next Generation of Ground Motion Attenuation (NGA) Project*. Earthquake Engineering Research Institute, Oakland, CA.

Somerville, P. G., Collins, N., Abrahamson, N., Graves, R., and Saikia, C. (2001). "Earthquake source scaling and ground motion attenuation relations for the Central and Eastern United States." *Final Report to the USGS under Contract 99HQGR0098*. USGS, Reston, VA.

Wells, D. L., and Coppersmith, K. J. (1994). "New empirical relationships among magnitude, rupture length, rupture width, rupture area, and surface displacement." *Bull. Seismol. Soc. of America* 84, 974–1002.

Working Group on California Earthquake Probabilities (WGCEP). (2013). "Uniform California earthquake rupture forecast, v. 3 (UCERF3)–The time-independent model." *USGS Open File Report 2013-1165*.

Zeng, Y., Anderson, J. G., and Yu, G. (1994). "A composite source model for computing synthetic strong ground motions." *Geophys. Res. Lett.*, 21, 725–728.